1. Objectives:

- To learn how to take input, creating functions and print the values in Python.
- To understand the usage of declaring variables, type casting, slicing and more.
- To know how to implement conditional statements if-else, while loops, and for loops in Python.

2. Introduction:

Python's a very easy language to start programing because it's a high-level programing language. In Python code is executed line by line by an interpreter. It makes easier for programmers to focus on problem-solving, building websites and even create machine learning stuff. Because it’s very easy to understand and works on almost any computer without needing to be changed.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** PyCharm Community Edition 20251.1.1

4. Experiment Implementation:

Experiment 1: Variables Declaration

Code:

```
age=24

name= ”Rezwan”

print(“Age:”, age)

print(“Name:”, name
```

Output:

```
Age: 24

Name: Rezwan
```

Experiment 2: User input in Python

Code:

```
num=int(input("Enter a number:"))

print("The number:",num)
```

Output:

```
Enter a number: 24

The number: 24
```

Experiment 3: Type Casting

Code:

```
a=10

b=10.5

Print(type(a))

Print(type(b))
```

Output:

```
<class ‘int’>

<class ‘float’>
```

Experiment No 4: Slicing

Code:

```
b = "Hello,World!"
print(b[2:])
```

Output:

```
llo,World!
```

Experiment 5: Operators

Code:

```
a = 10

b = 5

print("a+b =", a + b)

print("a*b =", a * b)
```

Output:

```
a+b = 15

a*b = 50
```

Experiment 6: Logical Operators

Code:

```
a=12
b=8
print(a>b or a<b)
print(a>b and a<b)
```

Output:

```
True

False
```

Experiment 7: Identity Operators

Code:

```
a=12
b=8
print(a is b)
print(a is not b)
```

Output:

```
True

False
```

Experiment 8: Membership Operators

Code:

```
numbers=[10,20,30,40,50]
print(20 in numbers)
print(100 in numbers)
```

Output:

```
True

False
```

Experiment 9: For Loop

Code:

```
i=1

for i in range(1,3,1):

    print(i)
```

Output:

```
1

2
```

Experiment 10: While loop

Code:

```
i=1
while i<5:
      if i==3:
         break
      print(i)
      i+=1
```

Output:

```
1

2
```

Experiment 11: if / elif / else

Code:

```
num1=18
num2=20
if num1>num2:
   print("Number 1 is greater than Number 2")
elif num1==num2:
     print("Number 1 is equal to Number 2")
else:
    print("Number 1 is less than Number 2")
```

Output:

```
Number 1 is less than Number 2
```

Experiment 12: Functions in Python

Code:

```
def multiply(a, b):

    return a * b

result = multiply(6, 7)

print("Multiplication:", result)
```

Output:

```
Multiplication: 42
```

5. Discussion & Conclusion:

In this lab, we explored the basic concepts of Python through practical examples. We learned how variables can be used without explicit declaration, practiced operators, and applied conditional statements (if-elif-else) to make logical decisions. Loops, both for and while, allowed us to repeat tasks efficiently, while functions showed their importance in making code reusable, organized, and easier to understand. Since Python is a high-level interpreted language with a simple, natural syntax, it felt beginner-friendly and easy to follow. By the end of the lab, we gained a clear foundation in Python programming that will support more advanced learning in the future.